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"Domain-Adaptive Deep Learning for Massive MIMO OFDM Channel Estimation Using Measured CSI"

Submitted by

Prabin Chandra Gautam (PUL080BEI028)

Saurab Gyawali (PUL080BEI039)

Smriti Khanal (PUL080BEI042)

A

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Declaration of Authorship

This is to certify that the minor project entitled *"Domain-Adaptive Deep Learning for Massive MIMO OFDM Channel Estimation Using Measured CSI"* submitted for the degree of Bachelor of Engineering in Electronics, Communication and Information Engineering comprises our original work and no part of this work has been used for the award of another course or degree. We declare that this work is our own. If any part of the work is not our own, we have indicated it by acknowledging the source of that part or those parts of the work with proper citations.

Name of the Student	**Signature**
Prabin Chandra Gautam	_____
Saurab Gyawali	_____
Smriti Khanal	_____

Date: _____

Approval

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The undersigned certify that they have read, and recommended to the Institute of Engineering for acceptance, a minor project entitled *"Domain-Adaptive Deep Learning for Massive MIMO OFDM Channel Estimation Using Measured CSI"* submitted by Prabin Chandra Gautam (PUL080BEI028), Saurab Gyawali (PUL080BEI039), and Smriti Khanal (PUL080BEI042) in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Electronics, Communication & Information Engineering.

Supervisor, *Dr. Nanda Bikram Adhikari*
Department of Electronics & Computer Engineering
Pulchowk Campus, IOE, TU

Internal Examiner,
Department of Electronics & Computer Engineering
Pulchowk Campus, IOE, TU

Head of Department, Prof. Dr. Ram Krishna Maharjan
Department of Electronics & Computer Engineering,
Pulchowk Campus, IOE, TU

Date: _____

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Department of Electronics & Computer Engineering
Pulchowk Campus
Pulchowk, Lalitpur
Nepal

Abstract

Channel state information (CSI) estimation is a fundamental component of Massive MIMO–OFDM systems that directly impacts beamforming, precoding, multi-user detection, and overall system throughput. Traditional estimators such as least squares (LS) and minimum mean square error (MMSE) are computationally expensive or require accurate channel statistics that may not be available in practice. Recent deep-learning approaches have shown promise but suffer from significant performance degradation when applied to real measured channels due to the simulation-to-real domain gap. This project presents a self-supervised domain-adaptive deep-learning framework for CSI estimation that bridges the gap between synthetic training data and real-world deployment. The framework integrates two complementary mechanisms: (i) a domain-aware loss function that embeds frequency correlation and spatial correlation priors into the training objective and (ii) a self-supervised domain adaptation module based on maximum mean discrepancy (MMD) and consistency regularisation to a residual CNN model.

The experimental evaluation uses both synthetic datasets generated via DeepMIMO ray-tracing model, and real over-the-air measured CSI obtained from the RENEW dataset. Results demonstrate that the proposed framework performs well on both synthetic and measured data, with the domain-adaptive loss providing the most significant improvement. In contrast, the inclusion of MMD and consistency losses provided limited additional benefit. These results highlight the effectiveness of domain adaptation in improving the transferability of the CNN-based channel estimator from synthetic to real-world measurements.

Keywords: Channel State Information, Massive MIMO, OFDM, Deep Learning, Domain Adaptation, Maximum Mean Discrepancy, CSI Estimation.

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Prabin Chandra Gautam (PUL080BEI028)
Saurab Gyawali (PUL080BEI039)
Smriti Khanal (PUL080BEI042)

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Abbreviations and Notation

5G	Fifth Generation
6G	Sixth Generation
AWGN	Additive White Gaussian Noise
BER	Bit Error Rate
CDL	Clustered Delay Line
CNN	Convolutional Neural Network
CSI	Channel State Information
DICHASUS	Distributed Channel Sounding User System
DLCS	Deep Learning Compressed Sensing
FDD	Frequency-Division Duplex
FFT	Fast Fourier Transform
GPU	Graphics Processing Unit
LDAMP	Learned Denoising-Based Approximate Message Passing
LMMSE	Linear Minimum Mean Square Error
LS	Least Squares
LSTM	Long Short-Term Memory
MIMO	Multiple-Input Multiple-Output
MMD	Maximum Mean Discrepancy
MMSE	Minimum Mean Square Error
MSE	Mean Square Error
NMSE	Normalised Mean Square Error
OFDM	Orthogonal Frequency-Division Multiplexing
QAM	Quadrature Amplitude Modulation
ResNet	Residual Network
RKHS	Reproducing Kernel Hilbert Space
SNR	Signal-to-Noise Ratio
TDD	Time-Division Duplex
TDL	Tapped Delay Line

Notation

Signals and Channel Model

$k,\ m$	Subcarrier index and receive-antenna index
N	FFT size (total number of subcarriers)
L	Maximum channel delay spread (number of taps)
n	Delay-tap index
M	Number of base-station antennas
$Y_{k,m}$	Received symbol on subcarrier k, antenna m
$X_{k,m}$	Transmitted (known) pilot symbol
$H_{k,m}$	Channel frequency response to be estimated
$h_{n,m}$	Delay-domain channel tap n at antenna m
$W_{k,m}$	Additive noise sample
σ_w^2	Noise variance
$\mathcal{CN}(0, \sigma_w^2)$	Circularly-symmetric complex Gaussian distribution
$\mathbf{H},\ \hat{\mathbf{H}}$	True and estimated channel matrices

Baseline Estimators

$\hat{H}_{\text{LS}}$	Least-squares (LS) channel estimate
$\hat{H}_{\text{MMSE}}$	Minimum-mean-square-error (MMSE) estimate
$Y_p,\ X_p$	Received and transmitted pilot symbols
P	Number of pilot subcarriers
$\mathbf{R}_H$	Channel covariance matrix $\mathbb{E}[\mathbf{h}\mathbf{h}^{\text{H}}]$
$\mathbf{I}$	Identity matrix

domain-aware Loss

$\mathscr{L}_{\text{total}}$	Total supervised loss on the source domain
$\mathscr{L}_{\text{MSE}}$	Reconstruction (mean-square-error) term
$\mathscr{L}_{\text{sparse}}$	frequency correlation (ℓ_1) term
$\mathscr{L}_{\text{corr}}$	spatial correlation term
$\lambda_{\text{sparse}},\ \lambda_{\text{corr}}$	Physics-prior weighting coefficients
N_s	Number of source (synthetic) samples
$\mathbf{H}^{(i)},\ \hat{\mathbf{H}}^{(i)}$	True and estimated channel for sample i
$\mathbf{R}_{\hat{H}}$	Empirical covariance of the estimated channel
$\mathscr{F}^{-1}\{\cdot\}$	Inverse FFT (frequency-to-delay transform)

Domain Adaptation

$\mathscr{L}_{\text{MMD}}$	Maximum-mean-discrepancy (MMD) loss
$\mathscr{L}_{\text{consistency}}$	Consistency-regularisation loss
$\mathscr{L}_{\text{final}}$	Overall combined training objective
λ_{MMD}, λ_{cons}	Adaptation weighting coefficients
$\mathscr{D}_S$, $\mathscr{D}_T$	Source and target feature distributions
$\mathbf{f}_S^{(i)}$, $\mathbf{f}_T^{(j)}$	Source and target feature embeddings
$\phi(\cdot)$	Feature map of the RKHS
$k(x, y)$	Gaussian radial-basis-function kernel
σ	RBF kernel bandwidth
n, m	Source and target batch sizes
N_t	Number of target samples
$\mathbf{x}_T^{(i)}$, $\tilde{\mathbf{x}}_T^{(i)}$	Target input and its perturbed version
$f(\cdot)$	Network prediction function
p	Pilot-masking probability

Operators and Norms

$\|\cdot\|_F$	Frobenius norm
$\|\cdot\|_1$, $\|\cdot\|_2$	ℓ_1 and ℓ_2 vector norms
$\|\cdot\|_{\mathscr{H}}$	Norm in the reproducing kernel Hilbert space
$(\cdot)^{\text{H}}$	Hermitian (conjugate) transpose
$\mathbb{E}[\cdot]$	Expectation operator

Chapter 1. Introduction

1.1. Background

Massive multiple-input multiple-output (Massive MIMO) combined with orthogonal frequency-division multiplexing (OFDM) forms the core air-interface technology of fifth-generation (5G) wireless systems and is widely regarded as a key enabler for sixth-generation (6G) networks. The deployment of large-scale antenna arrays at the base station, together with multicarrier transmission, provides substantial gains in spectral efficiency, spatial multiplexing, link reliability, and energy efficiency. However, these improvements are fundamentally dependent on the availability of accurate channel state information (CSI) at the receiver. CSI is required for coherent demodulation, beamforming, multi-user detection, link adaptation, and radio-resource allocation, and any inaccuracy in its estimate propagates directly into every downstream processing block of the transceiver chain.

Conventional channel estimation is performed by transmitting known pilot symbols on a subset of subcarriers and time slots, and then interpolating the channel response observed at those pilot positions to the entire time-frequency grid. The two classical estimators are least squares (LS), which simply divides the received pilot by the transmitted pilot and is therefore trivial to implement but extremely sensitive to noise, and minimum mean square error (MMSE), which incorporates the second-order statistics of the channel to suppress noise but requires accurate covariance knowledge and matrix inversions whose complexity scales unfavorably with the number of antennas. Both estimators degrade sharply under high mobility, sparse pilot grids, and large antenna configurations, which are precisely the operating regimes targeted by modern Massive MIMO systems.

In recent years, deep-learning-based channel estimators have attracted considerable attention because they are able to learn complex non-linear mappings directly from data and to exploit structural priors that are difficult to express in closed form. However, almost all such estimators reported in the literature are trained on synthetic channel data generated by analytical or ray-tracing models, and they suffer a substantial performance drop when applied to measured CSI because of the simulation-to-real domain gap. The work presented in this report addresses this generalisation problem by combining domain-aware learning, which embeds known wireless-channel structure into the training objective, with self-supervised domain adaptation, which transfers knowledge from synthetic to measured CSI without requiring labelled real data.

1.2. Problem Statement

Current deep learning-based CSI estimation methods suffer from several key limitations:

1. Heavy dependence on synthetic training datasets: Most models are trained on simulated channel data, which does not fully capture real-world propagation effects.
2. Poor generalisation to real measured environments: The domain gap between synthetic and real CSI leads to significant performance degradation when models are deployed in practical systems.
3. Lack of physical interpretability: Many deep learning models operate as black boxes and do not explicitly incorporate known wireless channel properties such as sparsity and spatial correlation.
4. Requirement of labelled channel measurements: Supervised learning approaches rely on ground-truth CSI, which is expensive and difficult to obtain in real deployments.

1.3. Objectives

1.3.1. Main Objective

The main objective of this project is to develop a channel-aware, domain-adaptive deep learning framework for Massive MIMO–OFDM channel state information (CSI) estimation, leveraging inherent wireless channel properties to enable robust generalization from simulated to real measured environments.

1.3.2. Specific Objectives

The specific objectives of the project are listed below:

1. To understand the characteristics and challenges of channel estimation in Massive MIMO–OFDM systems.
2. To explore deep learning techniques for improving domain generalization in channel estimation across diverse wireless channel environments.
3. To analyze the effectiveness of the developed framework using simulated and real-world wireless channel data.

1.4. Rationale

The rationale for combining domain-aware learning with self-supervised domain adaptation, rather than pursuing either approach in isolation, rests on the complementary nature of the two techniques. domain-aware learning embeds known wireless-channel characteristics in particular frequency correlation and spatial correlation across antenna elements directly into the training objective. This reduces the hypothesis space of the learned estimator and biases it towards solutions that are physically consistent, which in turn improves sample efficiency and reduces the dependence on large labelled training sets. However, domain-aware learning alone does not address the distribution shift between synthetic training data and real measured data; even a perfectly regularised model can still fail to generalise if the input statistics differ substantially from those seen during training.

Self-supervised domain adaptation, on the other hand, directly targets the distribution shift by aligning the feature representations of synthetic and real CSI without requiring labels in the target domain. Techniques such as maximum mean discrepancy (MMD) and consistency regularisation have been shown to be effective in computer-vision and speech-processing applications, and they are well suited to the wireless setting because they operate purely on internal feature embeddings and do not require any modification of the input pipeline.

By combining the two approaches, the framework developed in this project obtains the best of both worlds: the domain-aware loss ensures that the adapted model remains physically plausible, while the self-supervised adaptation step aligns the feature distributions of the two domains and narrows the simulation-to-real gap. This combination is particularly well suited to Massive MIMO–OFDM systems, where the underlying channel structure is well characterised and where the cost of obtaining labelled real CSI is prohibitive.

1.5. Scope

The scope of this project is defined as follows. The project considers Massive MIMO-OFDM channel estimation using uplink pilot observations. Both synthetic and measured channel data are considered. Synthetic channel data is generated using the DeepMIMO ray-tracing framework under an indoor propagation scenario, while real-world measurements are obtained from the Rice University RENEW indoor Massive MIMO dataset, consisting of a 64-antenna base station and seven user nodes, including six static users and one mobile user. The proposed CNN-based channel estimator is trained primarily on synthetic DeepMIMO data and evaluated on both synthetic and measured RENEW data to investigate synthetic-to-real generalisation. Conventional LS channel estimation

and a baseline CNN trained using MSE loss are used for comparison. The study further evaluates domain adaptation through an additional domain-adaptive loss and investigates the contribution of MMD and consistency-based loss components to improving transfer from the synthetic to the measured domain. Performance is evaluated primarily using normalised mean square error (NMSE), with particular emphasis on the generalisation gap between synthetic and real-world data and the improvement obtained through domain adaptation. Hardware-level impairments are not explicitly modelled in the synthetic data; however, their effects are inherently present in the measured RENEW dataset. The project is limited to channel estimation and does not address subsequent tasks such as beamforming, channel coding, or hardware implementation.

1.6. Limitations

The following limitations of the work are acknowledged. First, the synthetic dataset is generated using an idealised ray-tracing-based channel model and does not capture all propagation effects and imperfections present in real-world measurements, including hardware impairments, calibration errors, and other measurement-specific variations. Second, the current implementation focuses on a specific indoor OFDM configuration based on the DeepMIMO and RENEW datasets; extension to other carrier frequencies, bandwidths, and wideband system configurations requires further investigation. Third, real-time deployment considerations, such as inference latency, memory footprint, computational complexity, and hardware implementation, are outside the scope of this minor project and are left for future work.

1.7. Report Organisation

The remainder of this report is organised as follows. Chapter 2 reviews the related theory and the existing literature on channel estimation for Massive MIMO–OFDM systems, and identifies the gap that this project addresses. Chapter 3 describes the methodology of the proposed framework, including the research design, system architecture, mathematical modelling, algorithms, dataset preparation, experimental setup, evaluation metrics, and validation strategy. Chapter 4 provides the implementation details, covering the requirements specification, system overview, development environment, system implementation, interface and visualisation, deployment and execution, and the challenges encountered together with their solutions. Chapter 5 presents the results of the experimental evaluation and discusses their implications, including a comparative analysis against the baseline methods. Chapter 6 concludes the report and outlines recommendations for future work. The references and appendices follow.

Chapter 2. Literature Review

2.1. Theoretical Background

2.1.1. Massive MIMO–OFDM Systems

Massive MIMO refers to wireless systems in which the base station is equipped with a large number of antennas (typically 32 to hundreds) serving a relatively smaller number of user terminals simultaneously. The large antenna array provides three fundamental benefits: array gain through coherent combining, spatial multiplexing through multi-user beamforming, and interference reduction through spatial discrimination. When combined with OFDM, which converts frequency-selective fading channels into a set of parallel flat-fading subchannels, Massive MIMO–OFDM forms the foundation of modern cellular standards including LTE-Advanced and 5G NR.

In an uplink Massive MIMO–OFDM system with M base station antennas, K single-antenna users, and N OFDM subcarriers, the received signal at the base station on subcarrier k can be expressed as $\mathbf{y}_k = \mathbf{H}_k\mathbf{x}_k + \mathbf{w}_k$, where $\mathbf{H}_k \in \mathbb{C}^{M\times K}$ is the channel matrix, $\mathbf{x}_k \in \mathbb{C}^K$ is the transmitted symbol vector, and $\mathbf{w}_k \in \mathbb{C}^M$ is additive noise. Accurate knowledge of $\mathbf{H}_k$ across all subcarriers is essential for coherent detection and precoding.

2.1.2. Orthogonal Frequency Division Multiplexing

Orthogonal Frequency Division Multiplexing (OFDM) is a multicarrier modulation technique widely used in modern wireless communication systems. In OFDM, a high-rate data stream is divided into multiple lower-rate parallel streams, which are transmitted simultaneously over a set of closely spaced subcarriers. The subcarriers are designed to be mutually orthogonal, allowing their spectra to overlap without causing inter-carrier interference under ideal synchronization conditions. This improves spectral efficiency while simplifying equalisation in frequency-selective wireless channels.

An OFDM symbol is generated by mapping information symbols onto a set of subcarriers and applying an inverse Fast Fourier Transform (IFFT). At the receiver, the corresponding Fast Fourier Transform (FFT) is applied to recover the frequency-domain representation. A cyclic prefix (CP) is typically inserted at the beginning of each OFDM symbol before transmission. The CP is a copy of the end portion of the OFDM symbol and is used to reduce the effects of inter-symbol interference caused by multipath propagation.

If the duration of the useful OFDM symbol is T and the number of subcarriers is N, the

subcarrier spacing, Δf, is chosen such that

$$\Delta f = \frac{1}{T}. \tag{1}$$

This spacing ensures orthogonality between the subcarriers over the useful symbol duration. Consequently, although the individual subcarrier spectra overlap in frequency, the contribution of other subcarriers can be separated at the receiver when the signals are properly synchronised.

2.1.3. Classical Channel Estimation

Classical channel estimation techniques fall into two broad categories: pilot-based estimation and blind (or semi-blind) estimation. Pilot-based methods transmit known symbols at designated positions in the time-frequency grid and estimate the channel at those positions through comparison with the received signal. The LS estimator computes $\hat{H}_{\text{LS}} = Y_p / X_p$ at each pilot position independently, requiring no statistical knowledge but suffering from noise enhancement. The MMSE estimator incorporates the channel covariance matrix to suppress noise: $\hat{H}_{\text{MMSE}} = \mathbf{R}_H (\mathbf{R}_H + \sigma_w^2 \mathbf{I})^{-1} \hat{H}_{\text{LS}}$, achieving superior performance at the cost of computational complexity $O(M^3)$ for matrix inversion.

Blind methods exploit structural properties of the transmitted signal or the channel itself, such as cyclostationarity, subspace structure, or constant modulus properties, to estimate the channel without dedicated pilots. While spectrally efficient, blind methods typically require long observation windows and suffer from ambiguity problems that limit their practical applicability in mobile scenarios.

2.1.4. Deep Learning for Wireless Communications

Deep learning has emerged as a powerful paradigm for addressing problems in wireless communications where traditional model-based approaches face limitations. Neural networks can learn complex mappings directly from data, capture non-linear relationships that are difficult to express analytically, and adapt to varying channel conditions through training. In the context of channel estimation, deep learning approaches reformulate the problem as a supervised learning task in which a neural network learns to map received pilot observations (possibly corrupted by noise and interpolation artifacts) to the underlying channel response.

Convolutional neural networks (CNNs) are particularly well-suited to CSI estimation because the time-frequency channel response naturally exhibits a two-dimensional structure amenable to convolutional processing. Residual networks (ResNets) address the vanishing-gradient problem in deeper architectures through skip connections, en-

abling more accurate channel estimates. More recently, attention mechanisms and transformer architectures have been explored for capturing long-range dependencies in the channel response.

2.2. Related Work

2.2.1. Deep-Learning-Based Channel Estimation

Deep-learning approaches reformulate channel estimation as a supervised learning problem in which a neural network learns to map received pilot observations to the corresponding channel response [1, 6]. Convolutional neural networks (CNNs) exploit local correlations in the time–frequency plane through learned convolutional kernels and are therefore particularly well suited to capturing the structured patterns that characterise wireless channels [3]. Residual networks (ResNets) further improve training stability and depth through skip connections that allow gradients to flow directly to earlier layers. Compared with traditional estimators, these data-driven methods are able to capture complex non-linear relationships that are difficult to express in closed form, and they generally outperform classical estimators in challenging time-varying environments, provided that the training distribution is representative of the deployment distribution [16].

Recent studies have reformulated channel estimation as a structured-signal learning problem by treating the time–frequency channel response as a two-dimensional image and applying convolutional neural networks to exploit local correlations. ChannelNet, for instance, employs cascaded CNNs to refine coarse LS estimates into high-resolution channel reconstructions across the entire resource grid [3]. For millimetre-wave and Massive MIMO systems, deep learning has also been applied to hybrid analog-digital architectures; Elbir et al. [2] proposed deep neural networks for joint channel estimation and hybrid beamforming that directly map received pilot signals to beamforming matrices, while other works have explored beamspace channel estimation [4]. Other approaches, such as learned denoising-based approximate message passing (LDAMP) and deep-learning-based compressed sensing (DLCS), combine sparse signal recovery with neural-network-based denoising and precoding [9].

To model temporal channel variations, recurrent and attention-based architectures have been investigated. Gated recurrent unit (GRU) and long short-term memory (LSTM) networks capture channel dynamics without explicit statistical models [8], while attention mechanisms improve estimation accuracy by focusing computational resources on the most informative spatial and spectral regions [10]. Recent work has further explored alternative signal representations and end-to-end learning; wavelet-based approaches improve robustness to noise and interference [5], while some methods directly learn the

mapping from received signals to transmitted symbols without explicit channel estimation [6]. To support benchmarking and reproducibility, the DeepMIMO dataset provides a ray-tracing-based framework for generating realistic MIMO channels across diverse propagation environments [7], and Mohammed et al. [11] presented a comparative study of deep-learning channel estimation across multiple 5G channel models.

2.2.2. Domain Adaptation for Wireless Systems

Domain adaptation addresses the problem of distribution shift between training (source) and deployment (target) domains [18]. In the context of wireless communications, the source domain typically consists of synthetic channel data generated by analytical or ray-tracing models, while the target domain comprises real measured channels affected by hardware impairments, environmental factors, and unmodelled propagation effects [21]. Several approaches have been investigated for narrowing this simulation-to-real gap.

Unsupervised domain adaptation methods align feature distributions between source and target without requiring target-domain labels. Maximum mean discrepancy (MMD) measures the distance between the means of source and target samples in a reproducing kernel Hilbert space (RKHS) and provides a differentiable objective for minimising distribution divergence [20]. Adversarial methods employ a domain discriminator that tries to distinguish source from target features while the feature extractor tries to fool the discriminator, leading to domain-invariant representations [19]. Consistency regularisation enforces that the model produces similar outputs for perturbed versions of the same input, improving robustness to domain-specific variations.

Self-supervised methods leverage the intrinsic structure of the target data to provide supervision signals without manual labelling. In the context of CSI estimation, consistency can be enforced across time steps (temporal consistency), across adjacent subcarriers (frequency consistency), or across perturbed inputs (perturbation consistency). These methods have shown promise for adapting pre-trained models to new channel conditions without requiring ground-truth measurements.

2.2.3. Domain-Aware Machine Learning

Domain-aware machine learning incorporates domain knowledge from physical laws and constraints into the learning process, improving sample efficiency, interpretability, and generalisation. In the context of wireless communications, relevant physical priors include: (i) delay domain sparsity because signals arrive via a limited number of multipath reflections, each with a distinct time delay; (ii) spatial correlation across antenna elements due to limited angular spread; (iii) temporal correlation due to finite channel coherence time; and (iv) frequency correlation due to delay spread limitations [12–15].

These priors can be incorporated through physics-informed loss functions that penalise violations of expected physical structure, through architectural constraints that enforce desired properties, or through data augmentation strategies that generate physically plausible training examples. The approach taken in this project combines a physics-informed loss function with domain adaptation, leveraging both physical structure and distribution alignment to achieve robust cross-domain performance.

2.3. Gap Analysis

Based on the literature review, four principal gaps are identified in the existing body of work. These gaps are summarised in Table 1 and are discussed below.

Table 1. Gap Analysis of Existing Works

Existing Work	**Limitation**
CNN/ResNet-based CSI estimation	Primarily trained and evaluated on synthetic datasets, resulting in limited generalisation to real measured channels [1, 3].
MMSE and statistical estimators	Require accurate channel statistics and become computationally expensive in large-scale Massive MIMO systems [14, 15].
Deep-learning channel estimation	Often operates as a black box and does not explicitly exploit channel sparsity and spatial correlation [1, 16].
Domain-adaptation methods	Limited work addresses adaptation from simulated CSI to real measured CSI without labelled target-domain data [18–20].

Most existing CSI estimation frameworks rely heavily on simulated channel models and exhibit significant performance degradation when deployed in real-life environments. Furthermore, many deep-learning approaches lack physical interpretability and require large amounts of labelled data for training.

These limitations motivate the development of a domain-aware, self-supervised domain-adaptation framework capable of transferring knowledge from synthetic datasets to real measured CSI while preserving the underlying wireless-channel characteristics. The present project is designed to address precisely this gap: by integrating frequency correlation and spatial correlation priors into the training objective, and by performing unsupervised feature alignment between synthetic and measured CSI, the proposed framework aims to deliver estimators that are both accurate on simulated channels and robust on real ones.

Chapter 3. Methodology

3.1. Research Design

This project employs an experimental research methodology combining simulation-based analysis with real-world measurement validation. The research design follows a five-stage pipeline illustrated in Figure 1: (1) synthetic and measured CSI dataset preparation; (2) baseline estimator implementation; (3) domain-aware model training; (4) self-supervised domain adaptation; and (5) performance evaluation and analysis. Each stage builds upon the previous one, enabling systematic ablation studies to isolate the contribution of individual components.

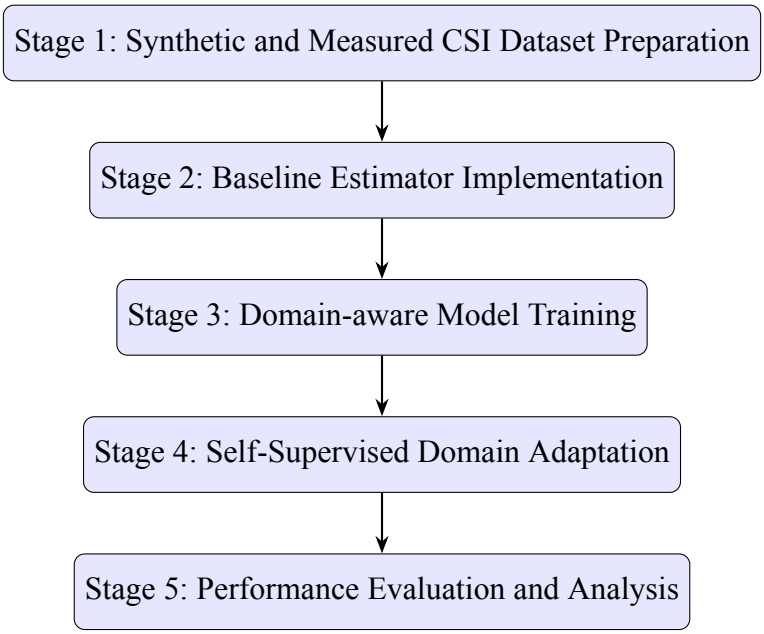


Figure 1. Research methodology pipeline showing the five stages of the project.

Stage 1: Synthetic and Measured CSI Dataset Preparation. Synthetic Massive MIMO-OFDM channel data is generated using the DeepMIMO ray-tracing framework under an indoor propagation scenario selected to provide a comparable environment to the measured RENEW dataset. The generated channel responses are used as ground-truth channel data for supervised training and evaluation. Real measured data is obtained from the RENEW indoor Massive MIMO dataset and processed to obtain frequency-domain pilot observations and channel estimates. The synthetic and measured datasets are maintained as separate source and target domains to evaluate the generalisation of the proposed estimator from synthetic to real-world measurements.

Stage 2: Pilot Processing and Conventional Baseline Estimation. The measured RENEW data is pre-processed according to its OFDM structure, including conversion of

interleaved I/Q samples into complex baseband samples, extraction of the pilot repetitions, FFT processing, and frequency-domain channel estimation. Conventional Least Squares (LS) and Minimum Mean Square Error (MMSE) channel estimators are implemented as classical baselines against the DL model. These estimators provide reference performance against which the learning-based channel estimation approaches are evaluated.

Stage 3: Residual CNN Model and Physics-Guided Training. A residual convolutional neural network (ResCNN) is developed for channel estimation, with the network learning the mapping between the received pilot observations and the corresponding channel response. The baseline ResCNN is trained using a normalized mean squared error (NMSE) loss on the synthetic DeepMIMO dataset. To incorporate known structural properties of wireless channels, additional spatial correlation and frequency correlation loss terms are introduced. These terms encourage the estimated channel to preserve the spatial and frequency-domain characteristics of the underlying channel response. The resulting domain-aware model is compared with the NMSE-only ResCNN baseline.

Stage 4: Domain-Adaptive Training and Additional Adaptation Losses. A domain-adaptive training strategy is incorporated to reduce the distribution mismatch between the synthetic DeepMIMO source domain and the measured RENEW target domain. The domain-adaptive loss is combined with the supervised channel-estimation objective during training. Additional adaptation mechanisms based on Maximum Mean Discrepancy (MMD) and consistency regularisation are also investigated to determine their contribution to synthetic-to-real generalisation. Since ground-truth channel labels are unavailable for the measured target-domain data, the domain adaptation components operate without requiring supervised target-domain channel labels.

Stage 5: Performance Evaluation and Analysis. The implemented channel estimation approaches are first trained and validated using the synthetic DeepMIMO dataset. This evaluation includes the conventional LS and MMSE estimators and two ResCNN models trained using different channel-estimation objectives, with NMSE used as the primary metric. The trained models are subsequently evaluated on the measured RENEW dataset to assess their transferability to real-world channel measurements. Since ground-truth channel responses are not available for the RENEW measurements, direct NMSE evaluation is not performed on the measured data. Instead, the real-world evaluation focuses on the adaptation gain achieved through the proposed domain-adaptive framework and the resulting BER, which provides an indirect assessment of the quality of the estimated channels in a communication setting.

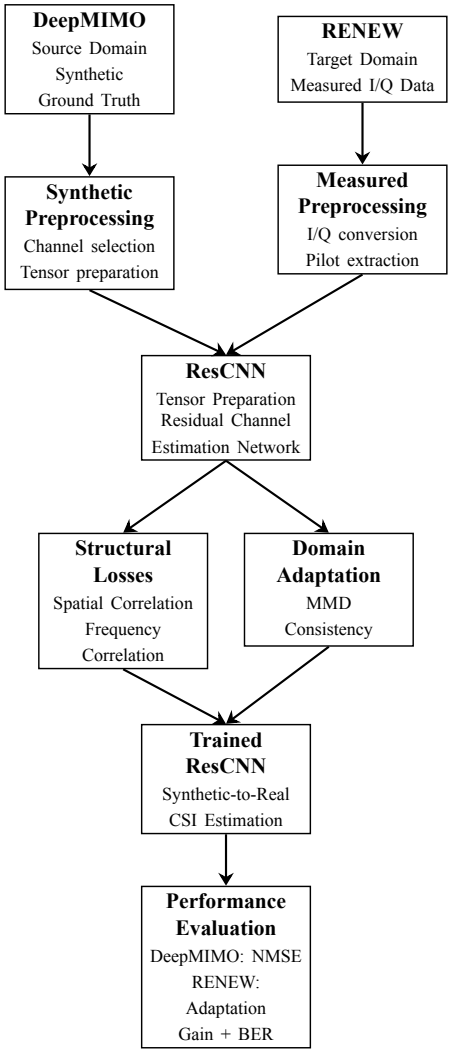


Figure 2. System architecture of the proposed synthetic-to-real CSI estimation and domain adaptation framework.

3.2. System Architecture

The proposed framework consists of four principal components: the synthetic source domain, the measured target domain, the ResCNN-based channel estimation network, and the training and adaptation framework. The implementation is carried out in Python, providing a common environment for data processing, channel estimation, neural network training, and evaluation.

The synthetic source domain is formed using DeepMIMO channel data, while the measured target domain is represented by the RENEW dataset. Both data sources undergo the required preprocessing to obtain compatible representations for the channel estimation network. The ResCNN forms the central estimation component and produces the estimated channel response from the processed pilot information.

During model training, the ResCNN is coupled with the channel-estimation and structural loss components, including the NMSE, spatial-correlation, and frequency-

correlation objectives. The domain adaptation component operates between the source and target domains to reduce their distribution mismatch. MMD and consistency-based mechanisms are incorporated as additional adaptation components and evaluated independently.

The resulting trained models are subsequently used for evaluation on the synthetic and measured domains. The overall architecture therefore provides a common framework for conventional channel estimation, residual CNN-based estimation, physics-guided training, and synthetic-to-real domain adaptation, while the detailed experimental procedure and evaluation strategy are described separately in the Research Design section.

3.3. Mathematical Modeling

This section presents the mathematical formulation of the proposed framework, covering the channel model, the baseline estimators, the domain-aware loss function, and the domain-adaptation objective. All notation is defined in the Abbreviations and Notation section.

3.3.1. OFDM and MIMO Channel Model

The discrete-time received signal on subcarrier k and antenna m is given by the fundamental equation:

$$\boxed{Y_{k,m} = H_{k,m} X_{k,m} + W_{k,m}} \tag{2}$$

where:

- $Y_{k,m}$ is the received complex symbol at subcarrier k and antenna m
- $H_{k,m}$ is the unknown channel frequency response to be estimated
- $X_{k,m}$ is the known transmitted pilot symbol
- $W_{k,m} \sim \mathscr{CN}(0, \sigma_w^2)$ is circularly symmetric complex Gaussian noise

The channel frequency response is related to the delay-domain representation $h_{n,m}$ through the discrete Fourier transform (DFT):

$$\boxed{H_{k,m} = \sum_{n=0}^{L-1} h_{n,m} e^{-j2\pi kn/N}} \tag{3}$$

where L is the maximum channel delay spread (number of taps) and N is the FFT size (total number of subcarriers). This relationship captures the fundamental duality between the frequency domain (where OFDM operates) and the delay domain (where the physical multipath structure manifests).

3.3.2. Baseline Estimators

The LS estimator at the pilot positions is the simplest approach, computing the ratio of received to transmitted pilot:

$$\boxed{\hat{H}_{\text{LS}} = \frac{Y_p}{X_p}} \tag{4}$$

While computationally efficient with complexity $O(P)$ for P pilots, the LS estimator treats each pilot independently and does not exploit correlations across time, frequency, or space, resulting in poor noise suppression.

The MMSE estimator that minimises the Bayesian mean square error incorporates the channel covariance matrix:

$$\boxed{\hat{H}_{\text{MMSE}} = \mathbf{R}_H \left(\mathbf{R}_H + \sigma_w^2 \mathbf{I}\right)^{-1} \hat{H}_{\text{LS}}} \tag{5}$$

where $\mathbf{R}_H = \mathbb{E}[\mathbf{h}\mathbf{h}^{\text{H}}]$ is the channel covariance matrix capturing spatial and frequency correlations. MMSE achieves optimal performance among linear estimators when the true covariance is known, but requires $O(M^3)$ operations for matrix inversion and suffers from performance degradation under covariance mismatch.

3.3.3. Domain-aware Loss Function

The training objective integrates data-driven reconstruction with wireless-channel structure priors. The total supervised loss on the synthetic source domain is defined as a weighted combination of three terms:

$$\boxed{\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{MSE}} + \lambda_{\text{sparse}} \mathscr{L}_{\text{sparse}} + \lambda_{\text{corr}} \mathscr{L}_{\text{corr}}} \tag{6}$$

where the individual components are defined as follows:

(1) Normalized Mean Square Error (NMSE) Term: Measures the reconstruction fidelity between estimated and ground-truth CSI, normalized by the power of the true channel:

$$\mathscr{L}_{\text{NMSE}} = \frac{1}{N_s} \sum_{i=1}^{N_s} \frac{\|\hat{\mathbf{H}}^{(i)} - \mathbf{H}^{(i)}\|_F^2}{\|\mathbf{H}^{(i)}\|_F^2 + \varepsilon} \tag{7}$$

where N_s is the number of source (synthetic) samples, $\hat{\mathbf{H}}^{(i)}$ is the estimated channel for sample i, $\mathbf{H}^{(i)}$ is the true channel, $\|\cdot\|_F$ denotes the Frobenius norm, and ε is a small constant for numerical stability.

(2) Frequency Correlation Term: Enforces the frequency correlation structure of the estimated channel to match that of the ground truth:

$$\mathscr{L}_{\text{FC}} = \frac{1}{N_s} \sum_{i=1}^{N_s} \left\| \mathbf{R}_{\hat{\mathbf{H}}}^{(i)} - \mathbf{R}_{\mathbf{H}}^{(i)} \right\|_F^2 \tag{8}$$

where $\mathbf{R} = \frac{1}{N_{ant}} \sum_{a=1}^{N_{ant}} \mathbf{h}_a \mathbf{h}_a^H \in \mathbb{C}^{N_{sc} \times N_{sc}}$ is the frequency correlation matrix computed across subcarriers, $\mathbf{h}_a$ is the channel vector for antenna a, and N_{sc} is the number of subcarriers.

(3) Spatial Correlation Term: Preserves the spatial correlation structure across antenna elements to maintain physically consistent Massive MIMO channel characteristics:

$$\mathscr{L}_{\text{corr}} = \left\| \mathbf{R}_{\hat{H}} - \mathbf{R}_H \right\|_F^2 \tag{9}$$

where $\mathbf{R}_{\hat{H}}$ and $\mathbf{R}_H$ are the empirical spatial covariance matrices of the estimated and true channel, respectively, computed across the antenna dimension.

The weighting coefficients λ_{sparse} and λ_{corr} control the relative influence of the two physical priors and are tuned on a held-out validation split. Through experimentation, we found optimal values of $\lambda_{\text{sparse}} = 0.01$ and $\lambda_{\text{corr}} = 0.1$.

3.3.4. Self-Supervised Domain Adaptation

For real measured CSI, adaptation is performed without requiring ground-truth labels using maximum mean discrepancy and consistency regularisation. This enables the model to adjust to the target domain distribution while maintaining the physical plausibility enforced by the domain-aware loss.

Maximum Mean Discrepancy (MMD): The squared MMD between the source feature distribution $\mathscr{D}_S$ and the target feature distribution $\mathscr{D}_T$ at an intermediate feature layer measures their distributional distance in a reproducing kernel Hilbert space (RKHS):

$$\boxed{\mathscr{L}_{\text{MMD}} = \left\| \frac{1}{n}\sum_{i=1}^{n} \phi(\mathbf{f}_S^{(i)}) - \frac{1}{m}\sum_{j=1}^{m} \phi(\mathbf{f}_T^{(j)}) \right\|_{\mathscr{H}}^2} \tag{10}$$

where:

- $\mathbf{f}_S^{(i)}$ and $\mathbf{f}_T^{(j)}$ are feature embeddings of source and target samples
- $\phi(\cdot)$ is the feature map of the RKHS (implicitly defined by the kernel)
- n and m are the numbers of source and target samples in a batch
- $\|\cdot\|_{\mathscr{H}}$ is the RKHS norm

We use a Gaussian radial-basis-function (RBF) kernel $k(x,y) = \exp(-\|x-y\|^2/2\sigma^2)$ with bandwidth σ set to the median pairwise distance between source and target embeddings.

Consistency Regularisation: Enforces that the network produces similar outputs for the same input under different perturbations:

$$\boxed{\mathscr{L}_{\text{consistency}} = \frac{1}{N_t}\sum_{i=1}^{N_t} \|f(\mathbf{x}_T^{(i)}) - f(\tilde{\mathbf{x}}_T^{(i)})\|_2^2} \tag{11}$$

where $\tilde{\mathbf{x}}_T$ is a perturbed version of the target input (obtained by adding Gaussian noise with $\sigma = 0.1$ or random pilot masking with probability $p = 0.1$), and $f(\cdot)$ denotes the network prediction function.

3.3.5. Final Combined Objective

The overall training objective of the proposed framework combines all loss terms into a unified optimisation target:

$$\boxed{\mathscr{L}_{\text{final}} = \underbrace{\mathscr{L}_{\text{MSE}}}_{\text{Reconstruction}} + \underbrace{\lambda_{\text{freq}}\mathscr{L}_{\text{freq}}}_{\text{Frequencycorrelation Prior}} + \underbrace{\lambda_{\text{spatial}}\mathscr{L}_{\text{spatial}}}_{\text{SpatialCorrelation Prior}} + \underbrace{\lambda_{\text{MMD}}\mathscr{L}_{\text{MMD}}}_{\text{Domain Alignment}} + \underbrace{\lambda_{\text{cons}}\mathscr{L}_{\text{consistency}}}_{\text{Robustness}}} \tag{12}$$

The complete objective thus balances reconstruction accuracy, physical structure preservation, and cross-domain alignment. At inference time, only the CNN forward pass is executed; the adaptation losses are disabled, resulting in zero overhead relative to the baseline model.

3.4. Algorithms Used

3.4.1. Baseline Deep-Learning Model

ResNet Variant (Residual CNN). This is a residual convolutional neural network designed to refine an initial channel estimate. Rather than directly predicting the true channel from scratch, which can be challenging for a plain CNN, this ResCNN learns only a residual correction to the linear least squares (LS) solution. The network starts from the LS estimate and progressively learns the mismatch between this estimate and the true channel. This residual learning strategy eases optimization because the network only needs to refine an already reasonable solution instead of learning the entire mapping. Skip connections within the architecture enable direct gradient flow, mitigating vanishing gradients and allowing stable training with deeper structures. Total parameters are approximately 0.48 million.

3.4.2. Domain-Aware Loss Implementation

The training loss combines three complementary components. The primary component is the normalized mean squared error. Two additional gradient-based terms act as regularizers: one enforces frequency correlation, and the other enforces spatial correlation across antennas. These three losses are weighted and summed to form the total training objective.

Algorithm 1 Domain-Aware Loss Computation

Require: Prediction $\hat{H}$, target H, frequency weight λ_{freq}, spatial weight $\lambda_{spatial}$
Ensure: Total loss $\mathscr{L}$ and individual loss components

1: Compute normalized mean squared error:$\mathscr{L}_{\text{NMSE}}$
2: Compute frequency-correlation loss along the subcarrier dimension: $\mathscr{L}_{\text{freq}}$
3: Compute spatial-correlation loss along the antenna dimension: $\mathscr{L}_{\text{spatial}}$
4: Combine the loss components:

$$\mathscr{L} = \mathscr{L}_{\text{NMSE}} + \lambda_{freq}\mathscr{L}_{freq} + \lambda_{spatial}\mathscr{L}_{spatial}$$

5: **return** $\mathscr{L}$, $\mathscr{L}_{\text{NMSE}}$, $\mathscr{L}_{freq}$, $\mathscr{L}_{spatial}$

3.4.3. Self-Supervised Domain Adaptation Algorithm

Algorithm 2 Unsupervised Domain Adaptation

Require: Pretrained source model g_{θ_0}, source dataset $\mathscr{D}_S$, target dataset $\mathscr{D}_T$
Ensure: Adapted model g_{θ^*}

Initialize the model with pretrained parameters $\theta \leftarrow \theta_0$
Freeze the input layers and residual feature-extraction blocks
Keep the output layer trainable for domain adaptation
for each training epoch **do**
    Sample a labelled source batch $(\mathbf{x}_S, \mathbf{y}_S)$
    Sample unlabelled target data $\mathbf{x}_T$
    Obtain source predictions and feature representations
    Obtain target feature representations and predictions from two perturbed views
    Compute supervised source estimation loss $\mathscr{L}_{\text{sup}}$
    Compute MMD loss $\mathscr{L}_{\text{MMD}}$ between source and target features
    Compute consistency loss $\mathscr{L}_{\text{cons}}$ between target predictions
    Gradually increase the weights of the adaptation losses during the initial training epochs
    Compute the total adaptation loss

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{sup}} + \lambda_{\text{MMD}}\mathscr{L}_{\text{MMD}} + \lambda_{\text{cons}}\mathscr{L}_{\text{cons}}$$

    Update the trainable parameters using AdamW
    Apply gradient clipping
    Evaluate the adapted model on the validation data
end for
Save the best-performing adapted model
return Adapted model g_{θ^*}

3.5. Dataset Preparation/Data Acquisition

3.5.1. DeepMIMO I3 2.4 GHz Scenario

The DeepMIMO I3 2.4 GHz scenario is an indoor ray-tracing-based wireless channel dataset designed for generating realistic MIMO channel responses. It provides multi-path propagation information including path power, phase, propagation delay, angle of arrival (AoA), angle of departure (AoD), and the corresponding interaction types along each propagation path. The scenario also provides transmitter and receiver spatial coordinates, enabling channel responses to be generated for different user locations. DeepMIMO supports the generation of frequency-domain MIMO channel matrices across configurable antenna arrays and OFDM subcarriers, making the scenario suitable for data-driven channel estimation and other wireless machine-learning applications. The underlying ray-tracing representation captures propagation phenomena such as line-of-sight propagation, reflection, diffraction, scattering, and transmission. [22, 23]

Table 2. Technical characteristics of the DeepMIMO I3 2.4 GHz scenario.

Characteristic	Description
Scenario	I3 indoor
Carrier frequency	2.4 GHz
Channel generation	3D ray tracing
Channel type	MIMO frequency-selective channel
Propagation information	Power, phase, delay, AoA, AoD
Spatial information	Transmitter and receiver coordinates
Propagation mechanisms	LoS, reflection, diffraction, scattering, transmission
Channel representation	Complex-valued MIMO channel response
Frequency-domain support	Configurable OFDM subcarriers
Antenna support	Configurable transmitter and receiver arrays
Primary application	Wireless communication and ML research

3.5.2. RENEW Indoor Measurement Dataset

The target-domain data is obtained from the RENEW Indoor Mobility Channel Measurement dataset released by Rice University. The dataset was collected using a 64-antenna RENEW software-defined massive MIMO base station and seven software-defined client nodes in a large indoor hall. Six clients were positioned statically at approximately 15 m from the base station, while the seventh client was mounted on a mobile robot. The measurement campaign includes both LoS and NLoS propagation conditions and different robot velocities. The uplink transmissions use time-orthogonal pilots based on the IEEE 802.11 Long Training Sequence (LTS) OFDM waveform, which contains 52 non-zero subcarriers [24, 25].

The RENEW data used in this work consists of two independent indoor measurement datasets. The first dataset contains measurements with the mobile user moving at 1 m/s and is used as the target domain during unsupervised domain adaptation. The second dataset contains measurements with the mobile user moving at 2 m/s and is reserved exclusively for final evaluation. The two datasets share the same measurement configuration and channel dimensions; they differ only in the mobile user speed and the number of available measurement samples.

The 1 m/s dataset contains 2001 recorded frames and is used during domain adaptation, while the independent 2 m/s dataset contains 3200 recorded frames and is used only for final evaluation. Thus, the 2 m/s measurements are not used during CNN pre-training or domain adaptation, providing an independent measured-domain test set for assessing generalisation.

Table 3. Common technical configuration of the RENEW datasets used in this work.

Parameter	Value
Dataset	RENEW Indoor Mobility Channel Measurements
Environment	Indoor
Carrier frequency	2.4 GHz
Base-station antennas	64
Number of users	7
Static users	6 NLoS
Mobile user	1 LoS
Pilot waveform	IEEE 802.11 LTS OFDM
FFT size	64
Cyclic prefix	16 samples
Occupied subcarriers	52
Null subcarriers	12
OFDM symbols per slot	10
Prefix samples	160
Postfix samples	160
Complex samples per slot	1120
Stored I/Q samples	2240
Channel representation	Complex-valued
Channel dimension	64×52

3.6. Experimental Setup

The experimental setup used to evaluate the proposed channel estimation framework is summarised in Table 4. The experiments use DeepMIMO I3 at 2.4 GHz as the synthetic source domain and the RENEW indoor measurement dataset as the real target domain. The synthetic data provides ground-truth channel responses for supervised training and validation, while the RENEW measurements are used for synthetic-to-real evaluation and unsupervised domain adaptation. All deep-learning models are implemented in Python using PyTorch.

3.7. Evaluation Metrics

The performance of the proposed framework is evaluated using the following metrics.

Normalised Mean Square Error (NMSE). The NMSE between the estimated channel $\hat{\mathbf{H}}$ and the true channel $\mathbf{H}$ is defined as:

$$\boxed{\text{NMSE} = \frac{\mathbb{E}\left[\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2\right]}{\mathbb{E}\left[\|\mathbf{H}\|_F^2\right]}} \tag{13}$$

expressed in decibels (dB) as $\text{NMSE}_{\text{dB}} = 10\log_{10}(\text{NMSE})$. Lower NMSE values indicate more accurate channel estimation.

Table 4. Experimental setup and principal configuration parameters.

Parameter	**Configuration**
Synthetic dataset	DeepMIMO I3 indoor scenario
Carrier frequency	2.4 GHz
BS antennas	64
Synthetic UE antennas	1
OFDM FFT size	64
Occupied subcarriers	52
Synthetic channel samples	105,250 user locations
Target dataset	RENEW indoor channel measurements
RENEW users	7 (6 static NLoS, 1 mobile LoS)
Pilot waveform	IEEE 802.11 LTS OFDM
Model parameters	483,906 trainable parameters
Source training loss	NMSE / Domain-aware loss
Domain adaptation	Supervised Learning, MMD, and consistency losses
Primary synthetic-domain metric	NMSE
Target-domain metrics	Adaptation gain, Generalization Gap and BER
Deep-learning framework	PyTorch

Measured-Data NMSE. Since the RENEW dataset does not provide ground-truth channel coefficients, the conventional NMSE cannot be computed directly. Instead, a self-consistency NMSE proxy is used. The ten repeated pilot transmissions are divided into two disjoint sets, producing two independent LS channel estimates, denoted by $\hat{H}_A$ and $\hat{H}_B$, corresponding to the same underlying channel realization. Each estimate is independently passed through the channel estimation network to obtain $\tilde{H}_A$ and $\tilde{H}_B$. The disagreement between these two predictions is used as an estimate of the residual channel-estimation error.

The measured-data NMSE proxy is defined as

$$\mathrm{NMSE}_{\mathrm{measured}} = \frac{\frac{1}{2}\mathbb{E}\left[\left\|\tilde{H}_A - \tilde{H}_B\right\|_F^2\right]}{\mathbb{E}\left[\left\|\frac{\tilde{H}_A+\tilde{H}_B}{2}\right\|_F^2\right]}. \tag{14}$$

The factor $1/2$ accounts for the fact that the difference between two independent estimates contains the error variance of both estimates. Thus, if $\tilde{H}_A = H + e_A$ and $\tilde{H}_B = H + e_B$, with independent errors of equal variance, then $\mathbb{E}\{\|e_A - e_B\|^2\} = 2\mathbb{E}\{\|e\|^2\}$. The denominator uses the power of the average prediction as a reference for the channel power. Therefore, this quantity provides a ground-truth-free estimate of channel-estimation error on the measured data. It is reported explicitly as an NMSE proxy rather than a conventional ground-truth NMSE.

Bit Error Rate (BER). The BER is evaluated on the independent 2 m/s RENEW dataset using QPSK-modulated symbols transmitted through the measured channels. For each channel, the received symbols are equalised using maximum-ratio combining (MRC) with the channel estimated by either the directly transferred or domain-adapted CNN. The equalised symbols are then demodulated using nearest-neighbour QPSK detection, and the decoded bits are compared with the known transmitted bits. In the final experiment, 100 independent measured channels are evaluated, with 10,000 QPSK symbols transmitted per channel, corresponding to a total of 104×10^6 transmitted bits. The BER therefore provides an end-to-end measure of the impact of channel estimation quality on symbol detection performance.

Generalisation Gap. The generalisation gap quantifies the performance difference when moving from the synthetic DeepMIMO domain to the measured RENEW domain. Since ground-truth channel coefficients are unavailable for RENEW, the measured-domain quantity is the self-consistency NMSE proxy described above:

$$\Delta_{\text{gen}} = \text{NMSE}_{\text{measured}} - \text{NMSE}_{\text{synthetic}}. \tag{15}$$

A smaller gap indicates better cross-domain generalisation.

Adaptation Gain. The adaptation gain measures the improvement obtained through unsupervised domain adaptation on the measured domain. It is calculated by comparing the direct-transfer model with the adapted model using the same measured-data NMSE proxy:

$$\Delta_{\text{adapt}} = \text{NMSE}_{\text{direct}} - \text{NMSE}_{\text{adapted}}. \tag{16}$$

A positive gain indicates that domain adaptation reduces the channel-estimation error on the measured data.

3.8. Validation Strategy

The validation strategy is designed to isolate the contribution of each component of the proposed framework. Three model variants are compared:

1. **Baseline ResCNN:** Trained only on synthetic data with standard NMSE loss
2. **Domain-aware Model:** Trained on synthetic data with domain-aware loss (MSE + sparsity + correlation)
3. **Full Proposed Framework:** Includes self-supervised adaptation on measured data

Chapter 4. Implementation Details

4.1. Requirements Specification

The implementation of the proposed framework is driven by a set of functional and non-functional requirements, which are summarised in Table 5 and Table 6 respectively.

Table 5. Functional Requirements

ID	Requirement
FR-1	**Data Processing:** Load and preprocess DeepMIMO and RENEW CSI data.
FR-2	**Baseline Estimation:** Implement LS and MMSE estimators.
FR-3	**ResCNN:** Implement the residual CNN channel estimator.
FR-4	**Domain-Aware Loss:** Implement NMSE, spatial-correlation, and frequency-correlation losses.
FR-5	**Domain Adaptation:** Adapt the model from DeepMIMO to RENEW data.
FR-6	**Adaptation Losses:** Implement MMD and consistency losses.
FR-7	**Training:** Train on labelled synthetic data and adapt using unlabelled measured data.
FR-8	**Evaluation:** Compute NMSE, BER, and adaptation gain.

Table 6. Non-Functional Requirements

ID	Requirement
NFR-1	**Accuracy:** Provide reliable channel-estimation performance.
NFR-2	**Generalisation:** Evaluate synthetic-to-real transfer performance.
NFR-3	**Reproducibility:** Use documented and repeatable experimental configurations.
NFR-4	**Modularity:** Allow individual estimation and adaptation components to be tested independently.
NFR-5	**Scalability:** Support batch-wise processing of large datasets.
NFR-6	**Maintainability:** Use structured and modular Python implementation.

4.2. System Overview

The implemented system follows the methodology described in Chapter 3. The mapping between the methodological stages and the implemented components is summarised below. Stage 1 (dataset preparation) is implemented using the DeepMIMO dataset generation pipeline for synthetic channel generation and the Rice RENEW dataset for real measured channel data. The DeepMIMO channels provide ground-truth CSI for supervised training and source-domain evaluation, while the RENEW measurements are

processed into the corresponding input representation for real-world evaluation and domain adaptation. Stage 2 (baseline channel estimation) includes the conventional LS and MMSE estimators, together with a baseline CNN trained using the standard NMSE loss. Stage 3 (domain-aware supervised learning) extends the CNN training objective by incorporating spatial and frequency correlation information to exploit the inherent structure of massive-MIMO channels. Stage 4 (synthetic-to-real domain adaptation) uses the pre-trained CNN and adapts it using unlabelled RENEW measurements through a combination of supervised source-domain loss, MMD-based domain alignment, and consistency regularisation. Stage 5 (evaluation) evaluates the resulting models on the synthetic DeepMIMO test data and an independent measured RENEW dataset using NMSE, the measured-domain self-consistency NMSE proxy, generalisation gap, adaptation gain, and end-to-end BER. The independent RENEW dataset is reserved exclusively for final evaluation and is not used during model adaptation or model selection.

4.3. Development Environment

The framework is implemented entirely in Python and executed across Kaggle/Colab and a local EndeavourOS environment. Kaggle was primarily used for computationally intensive model training and adaptation using an NVIDIA Tesla T4 GPU with 16 GB VRAM, while local experiments and data processing were performed on EndeavourOS. Package and library versions varied between the two environments; therefore, the specific versions are documented where required for reproducibility. DeepMIMO data generation, RENEW data processing, channel estimation, model training, domain adaptation, and evaluation were all performed using Python-based tools.

Table 7. Development Environment

Component	**Specification**
Programming Language	Python
Deep Learning Framework	PyTorch
Synthetic Data Generation	DeepMIMO Python implementation
Numerical Computing	NumPy, SciPy
Data Processing	h5py, NumPy
Visualization	Matplotlib
Primary Compute Platform	Kaggle and Colab Notebooks
Kaggle GPU	NVIDIA Tesla T4, 16 GB VRAM
Local Platform	EndeavourOS Linux
Execution Mode	GPU-accelerated and CPU-based experiments

4.4. System Implementation

This section describes the implementation details of each major software module, including the data structures, algorithms, and design decisions involved.

4.4.1. Data Generation and Loading Module

The data generation and loading module prepares the synthetic DeepMIMO source-domain data and the measured RENEW target-domain data for subsequent channel estimation and model training. Both datasets are processed using Python and converted into compatible complex-valued channel representations.

For the synthetic source domain, the DeepMIMO I3 indoor scenario at 2.4 GHz is generated using the Python implementation of DeepMIMO. The configuration uses a 64-antenna base station, a single-antenna user, and 64 OFDM subcarriers. The generated channel responses provide ground-truth CSI for supervised training and validation. A total of 105,250 user locations are used in the generated dataset. To emulate realistic noisy channel observations, the generated ground-truth CSI is used as the reference channel, and complex noise is introduced by randomly selecting the signal-to-noise ratio (SNR) for each sample from a range of 5-30 dB. This produces noisy channel observations under varying SNR conditions for training and evaluation of the proposed channel estimation models.

For the measured target domain, the RENEW indoor dataset is loaded from its native HDF5 format. The recorded interleaved I/Q samples are converted to complex baseband samples, followed by OFDM processing, pilot extraction, FFT processing, and selection of the 52 occupied subcarriers. The resulting channel representation contains 64 BS antennas and 52 frequency-domain subcarriers. The synthetic and measured datasets are maintained as separate source and target domains throughout the subsequent training and adaptation stages.

Table 8 summarises the principal datasets used in this work.

Table 8. Dataset Summary

Dataset	Source	Samples/Frames	Representation
DeepMIMO I3	Synthetic	105,250 users	Complex CSI
RENEW Indoor (1 m/s)	Measured	2,001 frames, 7 users	Complex I/Q data
RENEW Indoor (2 m/s)	Measured	3,200 frames, 7 users	Complex I/Q data

Data Separation

The experimental data separation is summarised as follows:

- **DeepMIMO:** used for supervised pre-training and source-domain testing.
- **Original RENEW dataset:** used exclusively for unsupervised domain adaptation.
- **Independent RENEW** $2\,\mathrm{m/s}$ **dataset:** used exclusively for final evaluation.

Consequently, the independent $2\,\mathrm{m/s}$ RENEW data was not used during model adaptation or model selection, providing a held-out measured-data evaluation of the adapted channel estimator.

4.4.2. Estimators Module

The channel estimation module implements LS, MMSE, and residual CNN (ResCNN) estimators. The LS and MMSE estimators provide conventional reference methods, while the ResCNN learns a residual correction to the LS channel estimate. Given B batch size, the ResCNN accepts a three-channel input consisting of the real and imaginary components of the LS estimate and the normalized SNR, with dimensions $(B, 3, 64, 52)$. It consists of an initial feature-extraction layer, six residual blocks, a feature-refinement layer, and an output layer producing the real and imaginary components of the estimated channel. Each residual block uses two 3×3 convolutional layers with GroupNorm, GELU activation, and dropout, with a skip connection between the block input and output. The network contains 483,906 trainable parameters and produces an output of dimensions $(B, 2, 64, 52)$.

Table 9. Architecture specifications of the residual CNN.

Specification	Value
Input shape	[Batch size, 3, 64, 52]
Input channels	3
Input features	$\Re(H_{LS})$, $\Im(H_{LS})$, normalized SNR
Output shape	[Batch size, 2, 64, 52]
Estimation strategy	Residual correction of LS estimate
Total convolutional layers	15

4.4.3. Loss Modules

4.4.4. Modules Design and Integration

The implementation is organised into five functional modules: data generation and loading, conventional channel estimation, deep-learning-based channel estimation, domain adaptation, and performance evaluation. The modules are implemented independently and integrated through common channel and pilot representations.

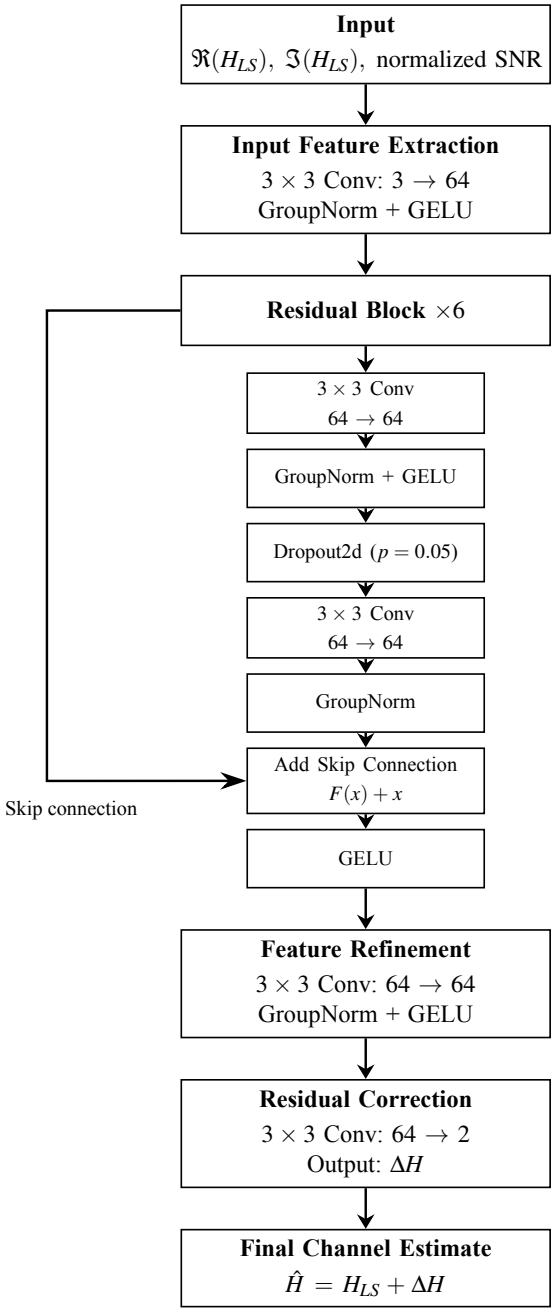


Figure 3. Architecture of the residual CNN for channel estimation.

Table 10. Components of the Domain-Aware Loss

Loss Component	Domain	Physical Property	Weight	Adjustment
NMSE	Overall	Reconstruction accuracy	1.0	Fixed
Frequency Gradient	Subcarrier	Frequency correlation	λ_{freq}	Fixed (CV-tuned, 0.05)
Spatial Gradient	Antenna	Spatial correlation	λ_{spatial}	Fixed (CV-tuned, 0.05)

Table 11. Implementation of Domain Adaptation Losses

Loss	Implementation	Domain	Weight
$\mathscr{L}_{\text{sup}}$	Standard MSE	DeepMIMO	1.0
$\mathscr{L}_{\text{MMD}}$	Unbiased multi-kernel Gaussian RBF MMD (5 kernels)	DeepMIMO + RENEW	$0 \leq \lambda_{\text{MMD}} \leq 0.5$
$\mathscr{L}_{\text{cons}}$	MSE between predictions from two disjoint 5-pilot views	RENEW	$0 \leq \lambda_{\text{cons}} \leq 0.1$

- **Data Generation and Processing:** Generates DeepMIMO channels, adds pilot noise, and obtains LS estimates for CNN input. Processes RENEW IQ data to obtain LS channel estimates from the repeated LTS pilots.
- **CNN Estimation:** Uses a residual CNN to refine the noisy LS channel estimate and recover the channel.
- **Domain Adaptation:** Pre-trains the CNN on labelled DeepMIMO data and adapts it to unlabeled RENEW data using MMD and consistency losses.
- **Independent Evaluation:** Evaluates direct and adapted models on an independent 2 m/s RENEW dataset using NMSE, generalisation gap, and adaptation gain.
- **BER Evaluation:** Transmits known QPSK symbols through the independent measured channels, performs MRC using the estimated channels, and computes BER from the detected bits.

4.4.5. Computational Complexity Analysis

The computational complexity of the three channel estimators is summarised in Table 12. Here, N_s is the number of channel samples, N_a is the number of BS antennas, N_f is the number of subcarriers, $H \times W$ is the spatial dimension of the CNN input, C_{in} and C_{out} are the input and output channels of a convolutional layer, K is the convolution kernel size, L is the number of convolutional layers, and P is the number of trainable model parameters.

Table 12. Computational complexity of the implemented estimators.

Estimator	Time Complexity	Space Complexity
LS	$\mathscr{O}(N_s N_a N_f)$	$\mathscr{O}(N_a N_f)$
MMSE	$\mathscr{O}(N_s N_f^2 + N_f^3)$	$\mathscr{O}(N_f^2)$
ResCNN	$\mathscr{O}\left(HW \sum_{l=1}^{L} K_l^2 C_l^{\text{in}} C_l^{\text{out}}\right)$	$\mathscr{O}(P + BC_{\max} HW)$

For the implemented ResCNN, $H = 64$, $W = 52$, $L = 15$, and $P = 483{,}906$, with B denoting the batch size.

Mathematically, LS is the most computationally efficient as it requires only element-wise multiplication, whereas MMSE involves matrix multiplications and inversion including the cubic matrix-inversion term. The ResCNN has a fixed complexity, where the spatial dimensions and convolutional layer dimensions are fixed after architecture selection; hence, unlike MMSE, its inference cost does not involve cubic matrix operations and remains predictable for every input.

4.5. Interface and Visualization

The implemented framework does not provide a standalone graphical or command-line interface, as the project is developed and evaluated primarily through Python-based Kaggle notebooks and Colab. The notebooks provide the interface for dataset loading, preprocessing, model training, domain adaptation, and evaluation, with intermediate results and metrics displayed during execution.

The visualisation component is used primarily for experimental analysis. Generated plots include training and validation loss curves, NMSE-versus-SNR comparisons on the synthetic dataset, BER-based comparisons on the measured dataset, and channel-response visualisations where applicable. These visualisations are used to assess model convergence, compare the implemented estimators, and analyse the effectiveness of domain adaptation. The final experimental results and corresponding figures are presented in Chapter 5.

4.6. Deployment and Execution

The complete source code is available at the project Google Drive repository referenced in the project submission. The repository contains the configuration files, the source code, the pre-trained checkpoints, the generated datasets, and the scripts required to reproduce every figure reported in Chapter 5.

4.7. Challenges and Solutions

Several challenges were encountered during implementation. The main challenges and solutions are summarised below.

Data format mismatch. DeepMIMO and RENEW use different data formats. A Python preprocessing pipeline was developed to convert RENEW interleaved I/Q samples into complex signals and a common tensor representation compatible with the synthetic data.

Memory limitations. The large RENEW measurements could not be processed entirely in memory. The data was therefore processed in smaller chunks to reduce memory usage.

Synthetic-to-real domain mismatch. The difference between DeepMIMO and measured RENEW channels limits direct transfer of the pretrained model. Domain-adaptive losses were therefore used to improve transfer without requiring ground-truth labels from RENEW.

Adaptation stability. MMD and consistency losses were gradually introduced during adaptation, while the early network layers and residual blocks were frozen to preserve the pretrained source-domain representation.

Chapter 5. Results and Discussion

5.1. Results

This section presents the experimental results obtained by evaluating the proposed framework on both the synthetic test set and the Rice RENEW measured test set. Four methods are compared throughout: LS, MMSE, the baseline CNN trained with the standard NMSE loss, and the proposed framework that combines domain-aware learning with self-supervised domain adaptation.

5.1.1. NMSE on Synthetic CSI

The results show a clear performance hierarchy across SNR. LS performs worst throughout, with NMSE decreasing from about −6 to −30 dB as SNR increases, but remaining substantially inferior because it directly amplifies noise. MMSE provides a strong improvement over LS, achieving approximately −16.8 dB at low SNR and reaching −38.7 dB at high SNR, demonstrating its ability to exploit channel statistics and suppress noise. The simple CNN trained with NMSE loss already outperforms both LS and MMSE in the low-to-moderate SNR region, starting around −22.98 dB and improving to −39.05 dB. However, the domain-aware CNN incorporating spatial and frequency correlation losses shows the strongest overall performance: it achieves lower NMSE than the simple CNN across the entire tested SNR range, improving from −22.98 dB at 5–7 dB SNR to −39.05 dB at 29–31 dB SNR. More importantly, compared with the corresponding simple CNN results, the domain-aware model provides a substantial improvement at higher SNRs, indicating that explicitly exploiting the spatial and frequency correlation structure of massive-MIMO channels improves generalization and channel reconstruction accuracy. Overall, the results demonstrate the progression LS < MMSE < simple CNN < domain-aware CNN, with the advantage of the learned estimators becoming increasingly evident as SNR increases.

5.1.2. Evaluation on an Independent RENEW Dataset

To evaluate whether the proposed domain adaptation method generalises beyond the data used during adaptation, an independent RENEW dataset recorded at a user velocity of $2\,\mathrm{m/s}$ was reserved exclusively for final evaluation. The original RENEW dataset was used during the unsupervised domain adaptation stage, whereas the $2\,\mathrm{m/s}$ dataset was not used for training, adaptation, or model selection.

Table 13 summarises the resulting performance.



Figure 4. Training of Simple ResCNN

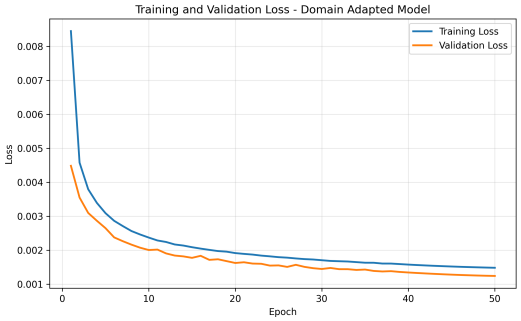


Figure 5. Training of Domain-aware CNN

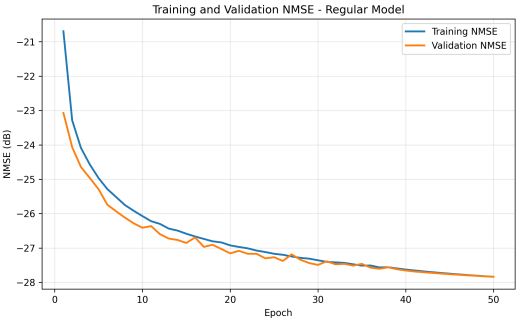


Figure 6. NMSE of Simple ResCNN during training

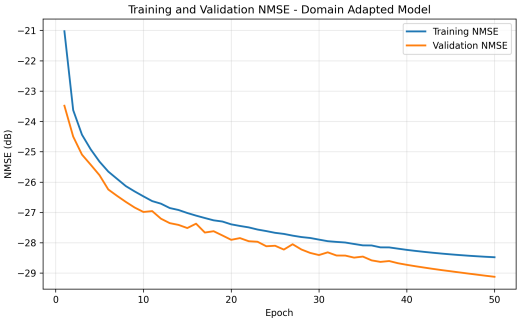


Figure 7. NMSE of Domain-aware CNN during training

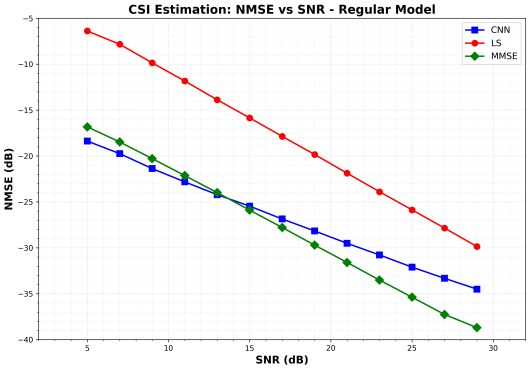


Figure 8. Comparison of Simple ResCNN with LS and MMSE

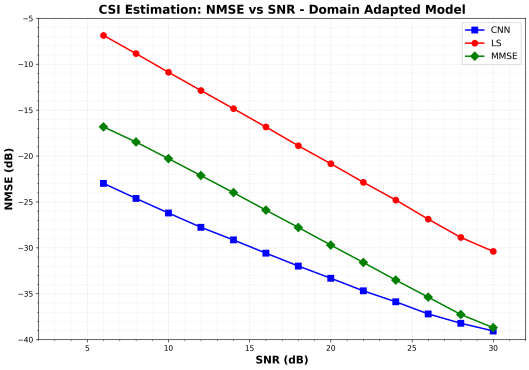


Figure 9. Comparison of Domain-Aware ResCNN with LS and MMSE

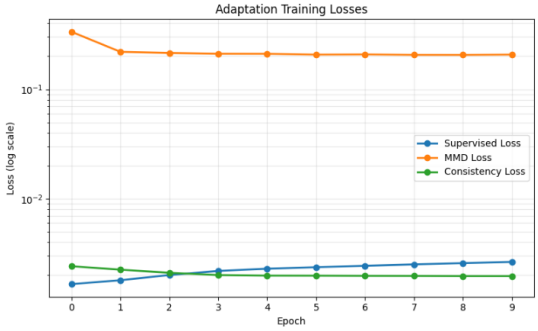


Figure 10. Domain Adaptation Training Process

Table 13. Performance on DeepMIMO and the independent RENEW $2\,\mathrm{m/s}$ ***test set. RENEW NMSE values represent a self-consistency NMSE proxy rather than ground-truth NMSE.***

Metric	**Direct**	**Adapted**
DeepMIMO NMSE (dB)	-28.7958	-28.5602
RENEW $2\,\mathrm{m/s}$ NMSE (dB)	-23.7056	-24.0603
Generalisation Gap (dB)	5.0902	4.4999

On the independent RENEW $2\,\mathrm{m/s}$ test set, the direct-transfer model achieves an NMSE of -23.7056 dB, whereas the adapted model achieves -24.0603 dB. This corresponds to an adaptation gain of

$$\Delta_{\text{adapt}} = -23.7056 - (-24.0603) = 0.3547 \text{ dB}. \tag{17}$$

Thus, domain adaptation improves the measured-domain NMSE by approximately 0.35 dB on data that was not observed during the adaptation process.

The generalisation gap also decreases from 5.0902 dB for the direct-transfer model to 4.4999 dB for the adapted model, giving a reduction of

$$\Delta_{\text{gap}} = 5.0902 - 4.4999 = 0.5903 \text{ dB}. \tag{18}$$

This reduction indicates improved transfer from the synthetic DeepMIMO domain to the measured RENEW domain. Importantly, the adapted model exhibits a small degradation on the DeepMIMO test set (from -28.7958 dB to -28.5602 dB), while achieving better performance on the independent measured dataset. This behaviour is consistent with domain adaptation, where a small amount of source-domain performance may be traded for improved target-domain generalisation.

Since the RENEW dataset does not provide ground-truth channel coefficients, the reported measured-domain NMSE is the self-consistency NMSE proxy described previously. Therefore, the observed improvement should be interpreted as evidence of improved self-consistent channel estimation on the independent measured data rather than as a conventional ground-truth NMSE improvement. Furthermore, given the relatively small improvement of approximately 0.35 dB, the observed gain may also be partially influenced by training randomness, parameter initialization, batch sampling, and measurement variability. Consequently, the results indicate a modest potential benefit from domain adaptation rather than a statistically conclusive improvement.

5.1.3. End-to-End BER Evaluation on Independent Measured Channels

We performed an end-to-end BER evaluation using real measured RENEW channels, generated QPSK symbols, transmitted them through $Y = HX + N$, equalised the received signals using the CNN-estimated channels, and compared the detected bits with the transmitted bits. The resulting BER performance is shown below.

Table 14. BER Performance Comparison: Direct CNN vs. Adapted CNN

SNR	Direct CNN BER	Adapted CNN BER
0 dB	2.81×10^{-4}	3.42×10^{-4}
5 dB	1.08×10^{-4}	7.34×10^{-5}
10 dB	2.87×10^{-5}	3.56×10^{-5}
15 dB	4.92×10^{-6}	2.15×10^{-6}

The end-to-end BER evaluation demonstrates the effectiveness of the CNN-based channel estimates on independent measured RENEW channels. As shown in Table 14, both models achieve consistently low BER across all evaluated SNR levels, with the Adapted CNN outperforming the Direct CNN at 5 dB and 15 dB, while the Direct CNN performs slightly better at 0 dB and 10 dB. This indicates that the original supervised CNN already provides effective channel estimates for the measured RENEW channels, while the additional adaptation offers limited benefit in terms of end-to-end BER performance under the evaluated conditions. Overall, the low BER values demonstrate that the CNN-estimated channels provide reliable channel information for subsequent signal detection over real measured RENEW channels.

5.2. Discussion

5.2.1. Effect of Domain-Aware Spatial and Frequency Correlation Learning

The domain-aware CNN demonstrates a substantial improvement over the conventional estimators and the standard CNN baseline, confirming the value of incorporating spatial and frequency correlation information into the learning objective. Overall, the domain-aware model achieves a maximum gain of approximately 12.84 dB over LS and 3.35 dB over MMSE, while improving upon the simple CNN by approximately 8.81 dB and 1.10 dB at the corresponding operating conditions. These gains indicate that the proposed correlation-aware losses provide information beyond conventional point-wise NMSE optimisation and allow the network to better exploit the inherent structure of massive-MIMO channels. The improvement over LS and MMSE highlights the advantage of learning-based estimation, while the additional gain over the simple CNN

demonstrates the specific contribution of the spatial and frequency correlation terms. This suggests that explicitly preserving dependencies across antenna and frequency dimensions leads to more effective channel representations and improved channel reconstruction, particularly in conditions where the underlying channel structure can be exploited more reliably.

5.2.2. Effect of Synthetic-to-Real Domain Adaptation

The second objective was to investigate whether unsupervised domain adaptation using MMD and consistency regularisation can improve the transfer of a CNN trained on synthetic DeepMIMO channels to real measured RENEW channels. The direct CNN already demonstrates strong transferability to the measured domain, indicating that supervised training on synthetic data is sufficient to learn channel representations that generalise to real measurements to a considerable extent. Applying the proposed adaptation produces a small improvement in the measured-domain self-consistency NMSE proxy, from -23.7056 dB to -24.0603 dB, while reducing the generalisation gap from 5.0902 dB to 4.4999 dB. The decrease in MMD and consistency losses during adaptation suggests that the training objective is able to modify the model towards greater target-domain consistency. However, these observations alone are not sufficient to conclude that the proposed adaptation is definitively effective. The measured-domain metric is a self-consistency NMSE proxy rather than a ground-truth NMSE, and the relatively small gain of 0.3547 dB may also be influenced by training randomness, measurement variability, and other stochastic effects. Furthermore, the end-to-end BER results do not show a consistent advantage for the adapted model, as it improves over the direct CNN at 5 dB and 15 dB but performs slightly worse at 0 dB and 10 dB. Therefore, based on the present experiments, the results provide evidence of a modest improvement in some target-domain metrics, but they are not sufficient to conclusively establish that MMD- and consistency-based adaptation improves synthetic-to-real channel estimation. Instead, the results suggest that the directly supervised CNN already provides strong synthetic-to-real transfer, while the additional adaptation should be regarded as a promising but inconclusive refinement that requires further validation, such as repeated training runs, statistical significance analysis, and evaluation across additional independent measured channels.

Chapter 6. Conclusions and Recommendations

6.1. Conclusions

6.2. Conclusion

This work investigated whether a CNN trained on synthetic channel data can be transferred to real measured massive-MIMO channels by incorporating domain-aware spatial and frequency correlation learning and unsupervised domain adaptation. The proposed framework first improves supervised CNN training by exploiting the inherent spatial and frequency correlations of massive-MIMO channels, followed by synthetic-to-real adaptation using supervised source loss, Maximum Mean Discrepancy (MMD), and consistency regularisation on unlabelled RENEW measurements. The synthetic-domain results demonstrate the effectiveness of the domain-aware learning approach, with the correlation-aware CNN providing substantial gains over conventional LS, MMSE, and the standard NMSE-trained CNN.

For the synthetic-to-real transfer problem, the directly supervised CNN already demonstrated strong performance on the independent measured RENEW data, showing that synthetic training can provide channel representations that transfer to real measurements despite the domain mismatch. The proposed MMD- and consistency-based adaptation produced a modest improvement in the measured-domain self-consistency NMSE proxy and reduced the generalisation gap. However, the improvement was small, and the adapted model did not consistently outperform the direct CNN in the end-to-end BER evaluation. Since the RENEW evaluation does not provide ground-truth channel coefficients and the observed improvements are relatively small, the present results cannot conclusively establish that the proposed adaptation method improves synthetic-to-real channel estimation. The observed gains may also be influenced by training randomness and measurement variability.

Overall, the study provides a qualified answer to the main research question. Domain-aware spatial and frequency correlation learning clearly improves the CNN's channel-estimation capability in the synthetic domain, while the effectiveness of the proposed unsupervised adaptation for synthetic-to-real transfer remains inconclusive under the evaluated conditions. The strong performance of the directly supervised CNN suggests that effective synthetic pre-training is already capable of producing transferable channel representations, whereas MMD and consistency-based adaptation may provide additional benefits but require more extensive validation to establish their reliability. Future work should therefore evaluate multiple adaptation runs, additional independent mea-

sured channels, and statistically robust performance comparisons to determine whether the observed adaptation gains are reproducible and significant.

6.3. Future Work

Several directions can be explored to extend the proposed domain-aware CSI estimation framework. First, *improving cross-domain and cross-scenario generalisation* is an important direction. The present study considers synthetic training data and real-world RENEW measurements; future work could investigate adaptation across different propagation environments, antenna configurations, user distributions, and hardware platforms to further assess the robustness of the proposed approach.

Transformer and hybrid architectures also provide a promising extension. While CNN and ResNet architectures effectively capture local spatial and frequency correlations, attention-based architectures could potentially model longer-range dependencies in the channel response. A systematic comparison of CNN, ResNet, and transformer-based estimators under the same domain-aware framework would therefore be valuable.

Joint pilot design and channel estimation is another promising direction. The current framework uses a fixed pilot structure, whereas jointly learning the pilot pattern and channel estimator could improve estimation accuracy under reduced pilot overhead. This would be particularly relevant for low-SNR and resource-constrained communication scenarios.

From a practical perspective, *real-time deployment and computational optimisation* should be investigated. Implementing the proposed estimator on software-defined radio or FPGA platforms would enable evaluation of inference latency, memory requirements, computational complexity, and energy consumption, providing a clearer assessment of its suitability for edge and real-time wireless systems.

Finally, the estimator could be evaluated in *downstream communication tasks*, such as beamforming, precoding, and link adaptation. Measuring the resulting throughput, spectral efficiency, and reliability would complement the current NMSE- and BER-based evaluation and provide a more comprehensive assessment of the practical benefits of improved CSI estimation.

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Appendix 1. Project Google Drive Link

The complete project artefacts, including source code, pre-trained model checkpoints, generated synthetic datasets, pre-processed Rice WARP dataset, configuration files, and scripts used to generate all figures reported in this document, are publicly available at:

`https://drive.google.com/drive/folders/1K1aGpx6om_VqkpZz5ud8mC3qCRRhjYfb`

Folder Organisation

The repository is structured into the following directories and files:

Directories

`Data_deepmimo/ :` Contains the DeepMIMO dataset files and related data processing resources.

`results/ :` Stores experimental results, model outputs, and performance metrics from the trained networks.

`results_without_loss/ :` Contains comparative results generated without the proposed loss function components.

All documents were last updated in August 2026, ensuring compatibility with the results reported in this report.